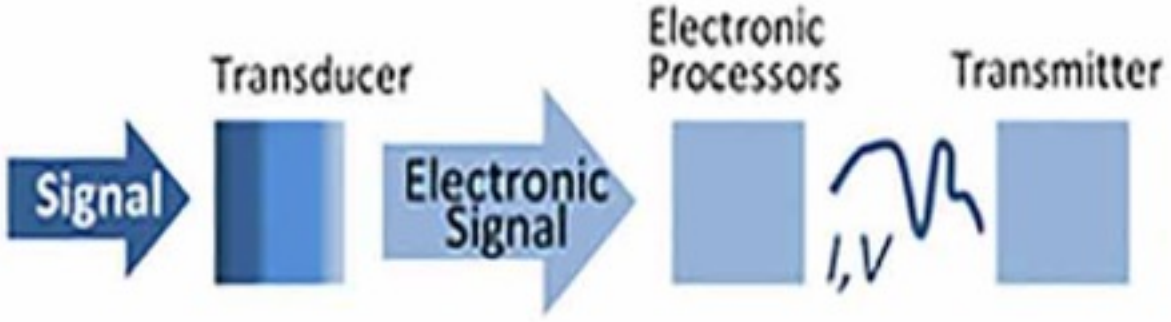
Antennas

The following subjects will be covered

- The Hertzian (point) Antenna
- The Dipole Antenna
- The Monopole Antenna.
- The Microstrip (patch) Antenna

Antennas

Communication Systems



Antenna



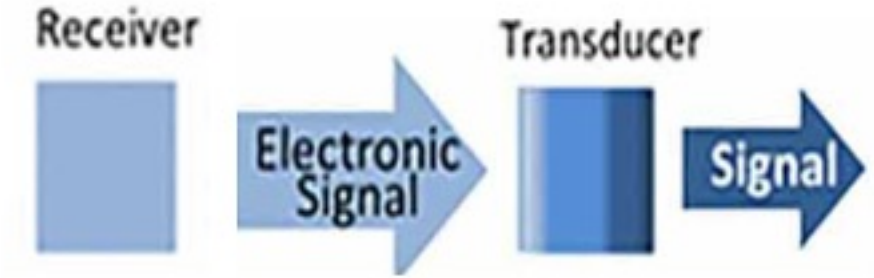
EM Waves



EM Waves







Antenna

The Hertzian Dipole

This is a basic antenna element, a very short wire carrying current, or a differential antenna (a point source).

Basic assumptions:

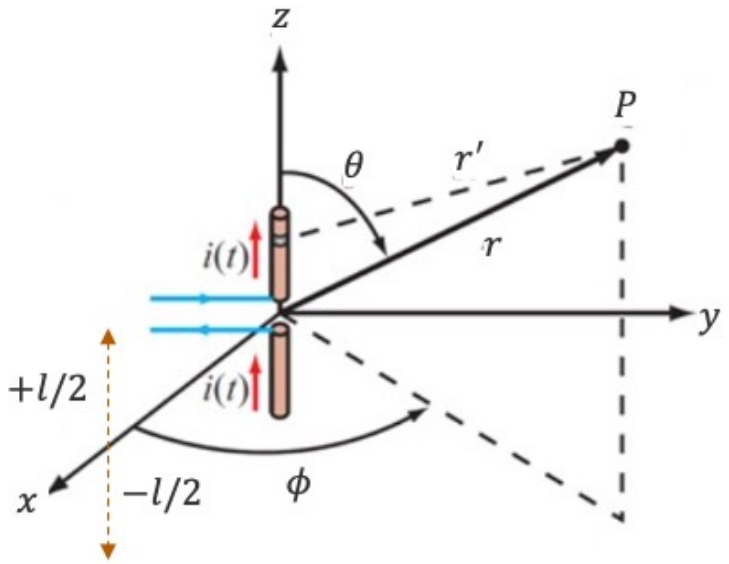
1) Infinitesimal length compared to distances involved (the wavelength)

$$l \ll \lambda \qquad l < \lambda/50$$

2) The current along the wire becomes constant.

$$I(z) = I_o \cos \omega t \approx I_o$$

3) The radiated waves are spherical.



r: Range ϕ: Azimuthal θ: Zenith

Spherical coordinates simplify the work

$$\vec{E} = E_r \vec{a}_r + E_\phi \vec{a}_\phi + E_\theta \vec{a}_\theta$$

$$\vec{H} = H_r \vec{a}_r + H_\phi \vec{a}_\phi + H_\theta \vec{a}_\theta$$

Because the current flows in the z-direction, the resulting fields have the following components

$$E_r, E_\theta, H_\phi$$

$$E_\phi = H_\theta = H_r = 0$$

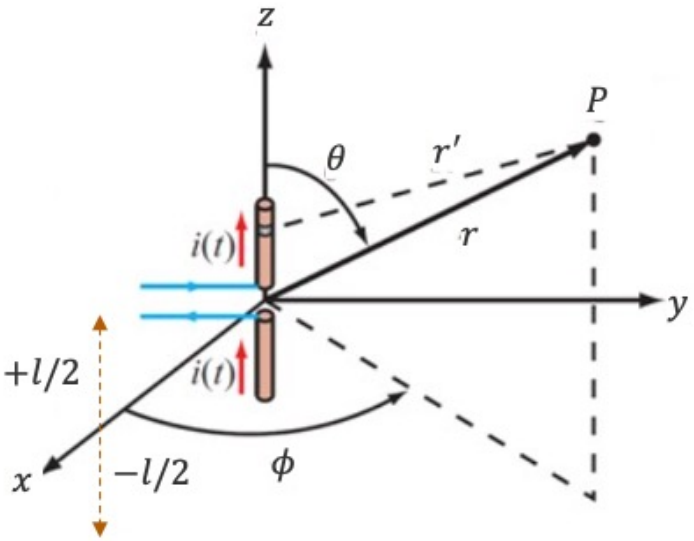
The Hertzian Dipole

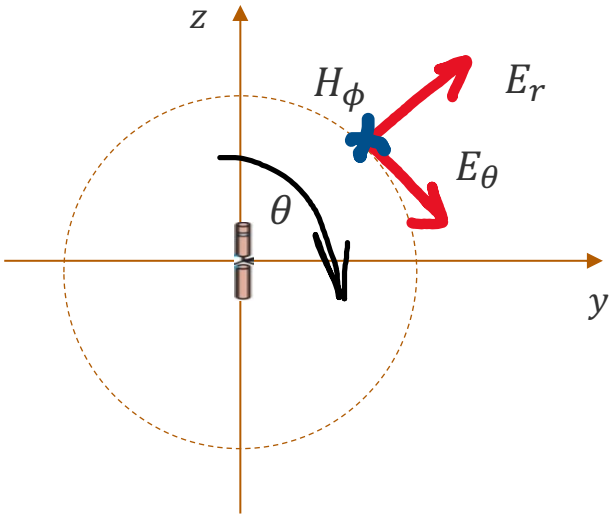
The existing field are as follows.

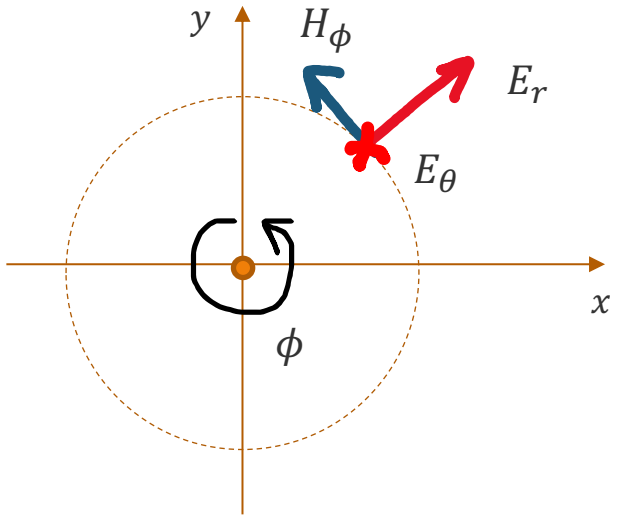
$$E_r = \frac{2I_o\, l\, \beta^2}{4\pi}\eta_o \left[\frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\cos\theta\; e^{-j\beta r}$$

$$E_\theta = \frac{I_o\, l\, \beta^2}{4\pi}\eta_o \left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2} - \frac{j}{(\beta r)^3}\right]\sin\theta\; e^{-j\beta r}$$

$$H_\phi = \frac{I_o\, l\, \beta^2}{4\pi}\left[\frac{j}{\beta r} + \frac{1}{(\beta r)^2}\right]\sin\theta\; e^{-j\beta r}$$







The Hertzian Dipole

Far-field Approximation.

r very large $\quad \beta r \gg 1$

$$\frac{2\pi}{\lambda} r \gg 1 \rightarrow r \gg \frac{\lambda}{2\pi}$$ Far-field Zone

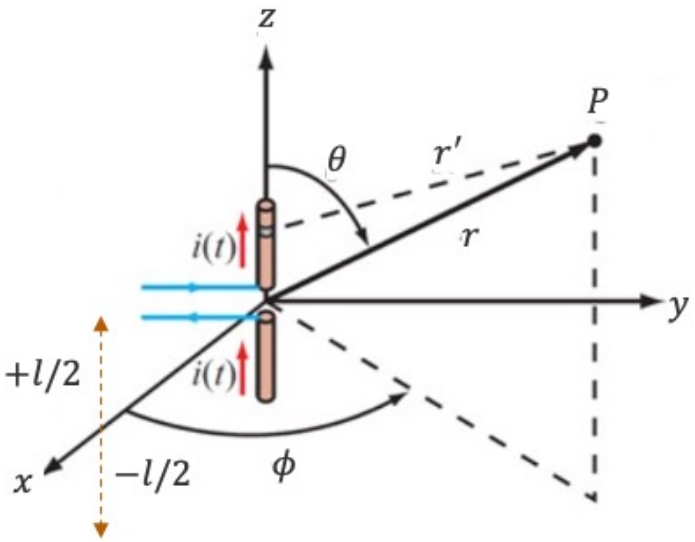
$$E_r \approx 0$$

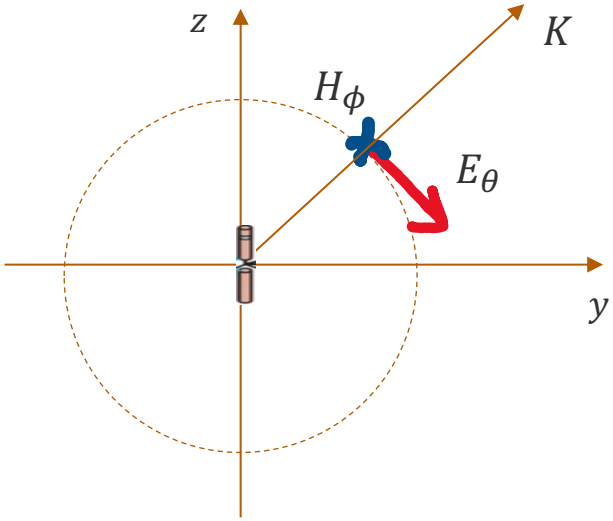
$$E_\theta = \frac{jI_o\, l\, \beta}{4\pi} \eta_o \sin\theta \left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = \frac{jI_o\, l\, \beta}{4\pi} \sin\theta \left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = E_\theta / \eta_o$$

Plane-wave behavior





The Hertzian Dipole

Time-average Power density

$$\vec{S}_{ave} = \frac{1}{2} Re(\vec{E} \times \vec{H}^*)$$

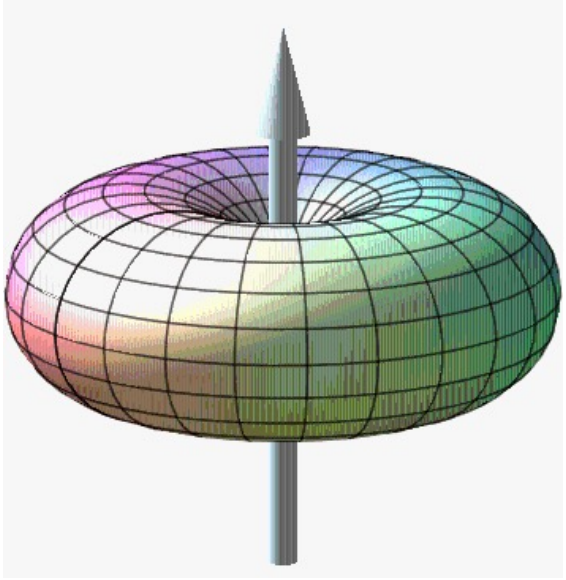
$$\vec{S}_{ave} = \frac{\eta_o \beta^2 I_o^2 l^2}{32 \pi^2 r^2} (\sin\theta)^2 \ \vec{a}_r$$

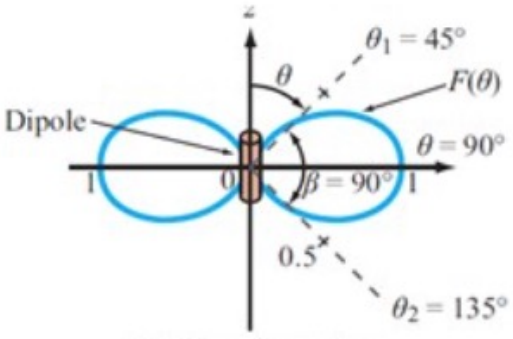
$$\vec{S}_{ave} = \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2 (\sin\theta)^2 \ \vec{a}_r$$

$$\vec{S}_{ave} = S_o(r) \ (\sin\theta)^2 \ \vec{a}_r$$

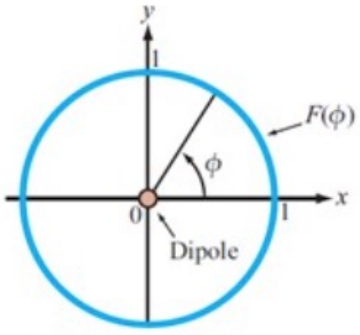
$$S_o(r) = \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2$$

$$S_{ave_max} = S_o(r)$$





Elevation plane $\phi = 0$



Azimuthal plane $\theta = \frac{\pi}{2}$

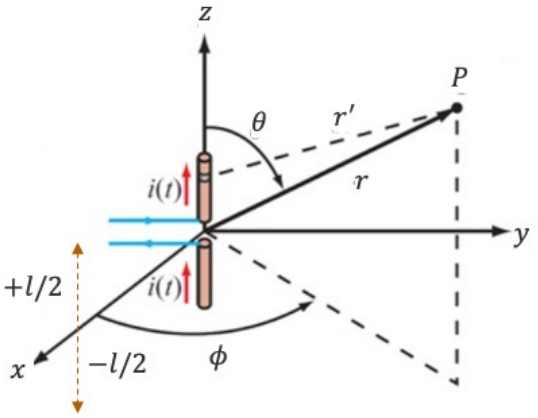
The Hertzian Dipole

Radiated Power

$$P_{rad} = \oint_S \vec{S}_{ave} \cdot \vec{d}_S$$

$$P_{rad} = \int_0^{2\pi} \int_0^{\pi} \frac{\eta_o I_o^2}{8r^2} \left(\frac{l}{\lambda}\right)^2 (\sin\theta)^2 r^2 \sin\theta \, d\theta d\phi$$

$$P_{rad} = \frac{\pi \, \eta_o I_o^2}{3} \left(\frac{l}{\lambda}\right)^2$$



Radiation Resistance

$$P_{rad} = I_{rms}^2 \, R_{rad}$$

$$P_{rad} = \frac{1}{2} I_o^2 \, R_{rad}$$

$$R_{rad} = 80\,\pi^2 \left(\frac{l}{\lambda}\right)^2 \approx 1\ \Omega$$

Example

The maximum signal picked from a radio station 2 km away has a magnetic field magnitude of 5 μA/m. Find the station radiated power. $l/\lambda = 1/25$.

$$|H_\phi| = \frac{I_o \, l \, \beta}{4\pi r} \sin\theta \ = 5\mu \rightarrow I_o = 0.5\ A$$

$$P_{rad} = \frac{\pi \, \eta_o I_o^2}{3} \left(\frac{l}{\lambda}\right)^2 = 157.7\ mW$$

$$R_{rad} = 80\,\pi^2 \left(\frac{l}{\lambda}\right)^2 = 1.26\ \Omega$$

The Hertzian Dipole

Efficiency

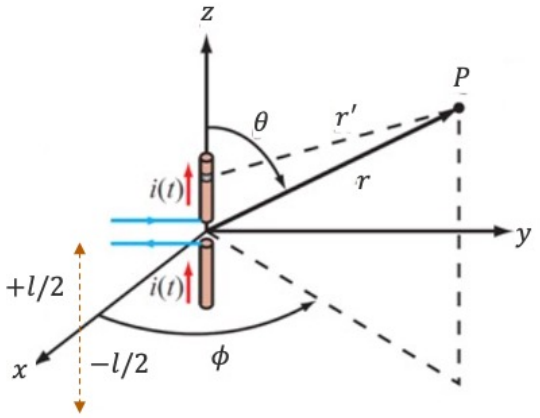
$$\eta = \frac{P_{rad}}{P_{tot}} = \frac{P_{rad}}{P_{rad} + P_{loss}}$$

Gain

$$G = \frac{S_o}{P_{tot}/4\pi r^2}$$

Directivity

$$D = \frac{S_o}{P_{rad}/4\pi r^2} \approx 1.5$$



Example: A 1-m dipole is excited by a 5-MHz current of amplitude 5 A. Find the broadside power density at 2 km.

Check if it's a Hertzian dipole. $\lambda = \frac{v}{f} = 3\times10^8/5\times10^6 = 60\ m$ $\qquad \frac{l}{\lambda} = \frac{1}{60}$

$$S_{ave} = \frac{\eta_o I_0^2}{8(2000)^2}\left(\frac{l}{\lambda}\right)^2 (\sin 90°)^2 = 8.2\times10^{-8}\ \text{W/m}^2$$

The Dipole Wire Antenna





The Dipole Wire Antenna

The dipole antenna length is $\lambda/2$, or any multiple, which is the resonance condition.

The analysis follow the same steps as the Hertzian dipole with one very important difference: the current distribution along the antenna is not constant.

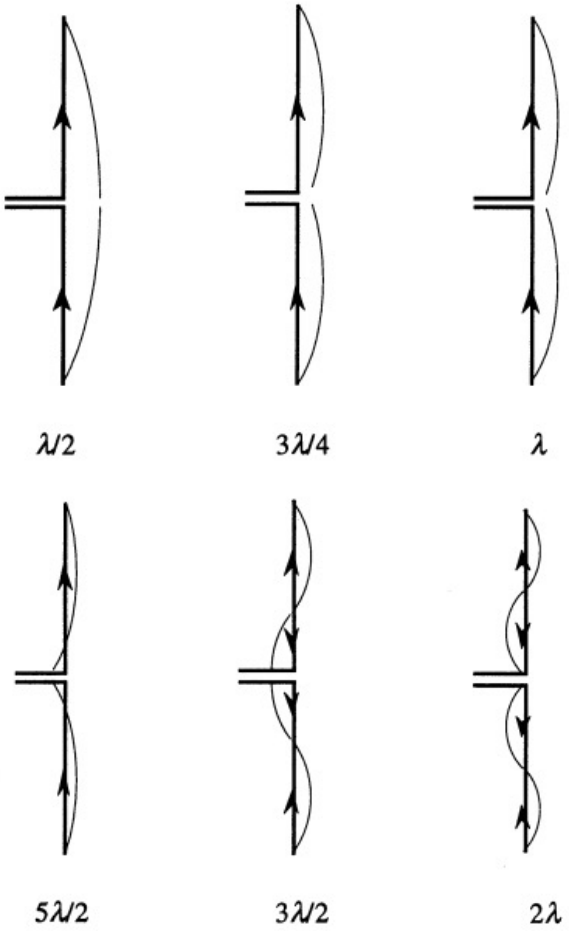
The current along the wire is

$$I(z) = I_o \cos \beta z \; = I_o \cos \frac{2\pi}{\lambda} z$$

Far-field Approximation.

$$E_\theta = j60\, I_o \frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\left[\frac{e^{-j\beta r}}{r}\right]$$

$$H_\phi = \frac{E_\theta}{\eta_o} = \frac{jI_o}{2\pi}\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\left[\frac{e^{-j\beta r}}{r}\right]$$



The Dipole Wire Antenna

Power density

$$\vec{S}_{ave} = \frac{\eta_o I_o^2}{8\pi^2 r^2}\left(\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right)^2 \vec{a}_r$$

$$\vec{S}_{ave} = S_o(r)\left(\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right)^2 \vec{a}_r$$

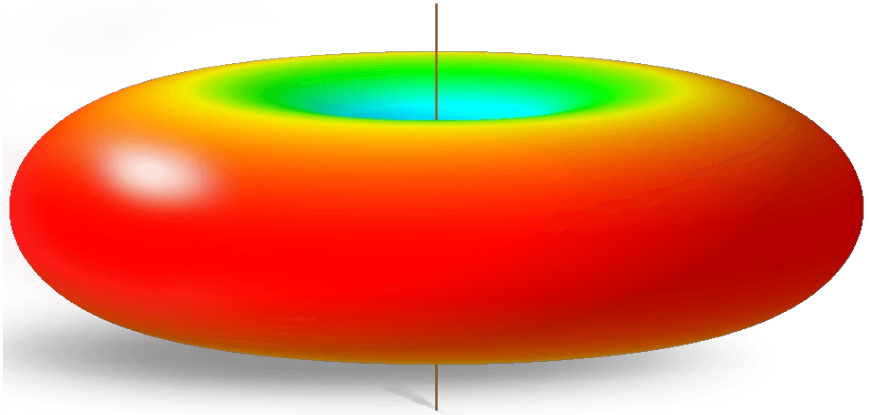
$$S_o(r) = \frac{15\, I_o^2}{\pi\, r^2}$$

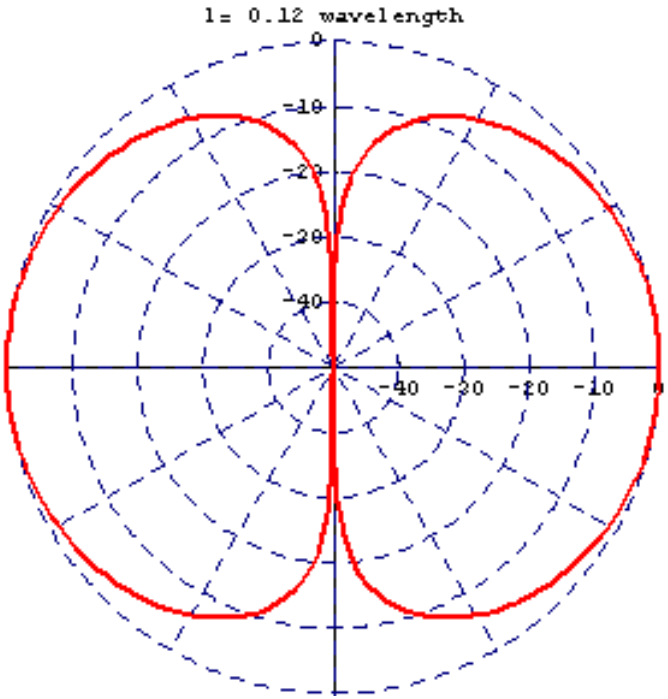
Radiated Power

$$P_{rad} = \oint_S \vec{S}_{ave} \cdot \vec{d}_S \approx 36.56\, I_o^2$$

$$R_{rad} \approx 73\ \Omega$$

$$D \approx 1.64$$





Example

The maximum signal picked from a radio station 2 km away has a magnetic field magnitude of 5 μA/m. The station uses a center-fed dipole antenna. Find the station radiated power.

$$|H_\phi| = \frac{I_o}{2\pi r} = 5\mu \quad \rightarrow \quad I_o = 0.063\ A$$

$$P_{rad} = (36.57)(0.063)^2 = 144\ mW$$

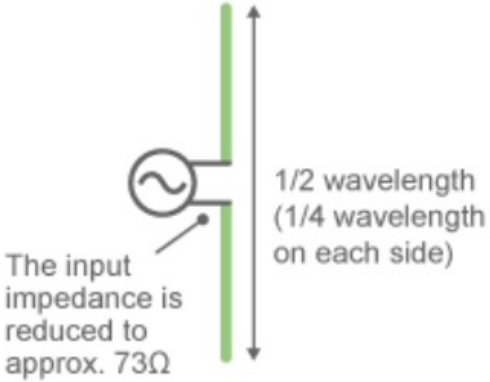
The mono-pole wire antenna

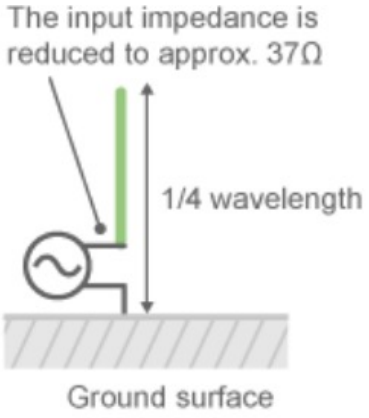
$$l = \lambda/4$$

$$P_{rad} \approx 18.28\ I_o^2$$

$$R_{rad} \approx 36.5\ \Omega$$

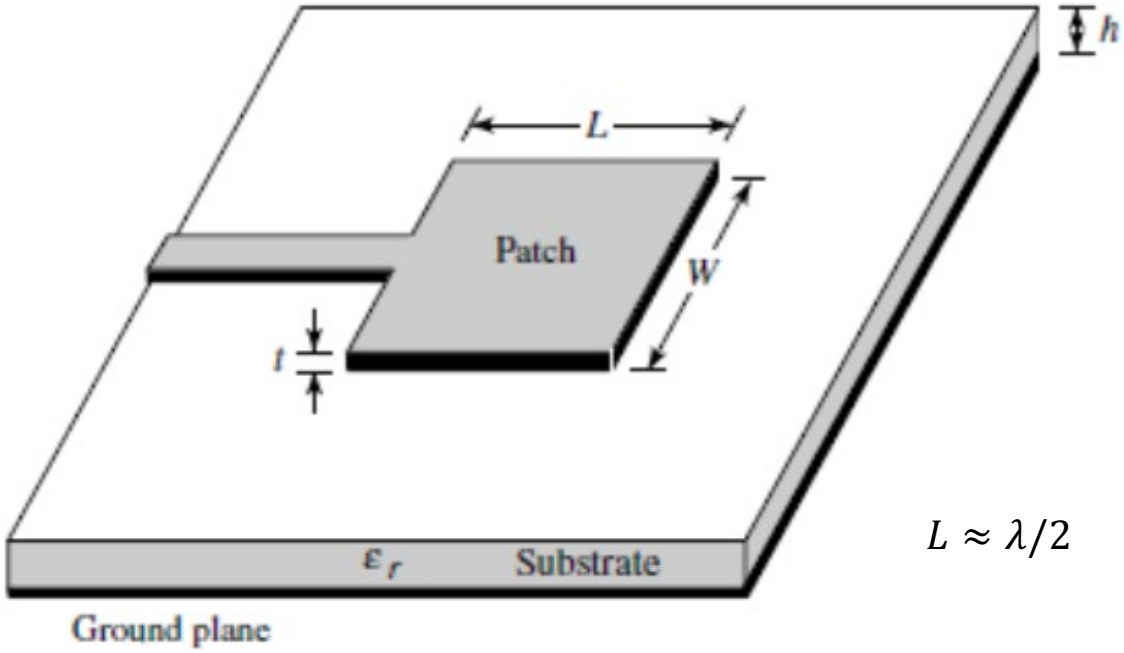
$$0 \le \theta \le \frac{\pi}{2}$$



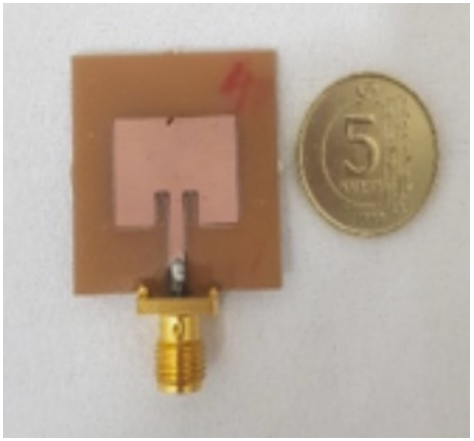


The Microstrip (patch) Antenna

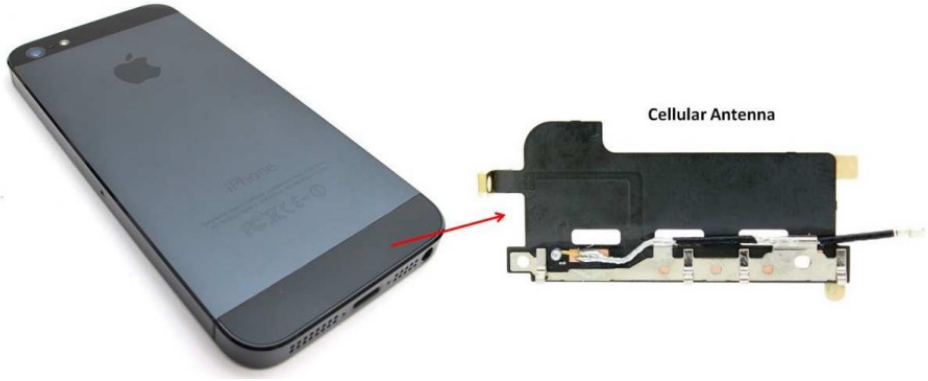
The Patch antenna is one of the half-wavelength resonating structures ($\lambda/2$, for the fundamental mode). Since its inception in the 1970's, it has become, by far, the most popular and versatile antenna structure for the microwave frequency range.



$L \approx \lambda/2$

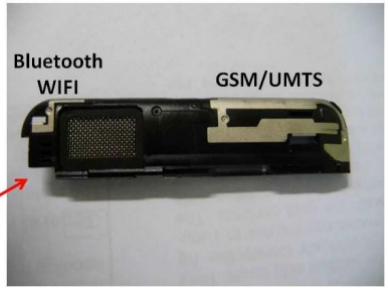


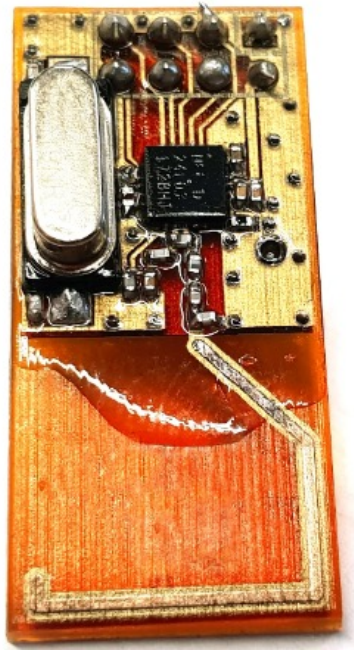
The Microstrip (patch) Antenna

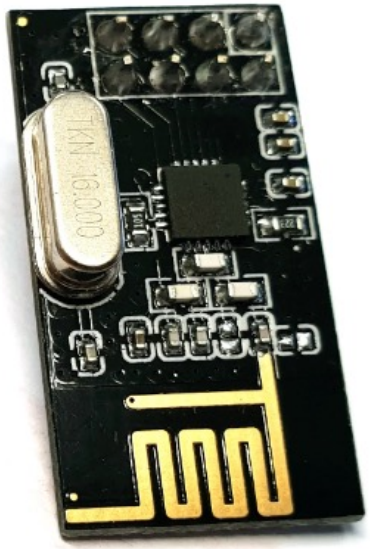


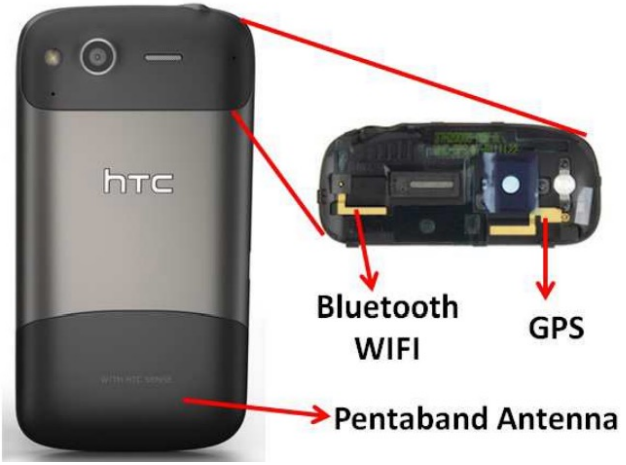






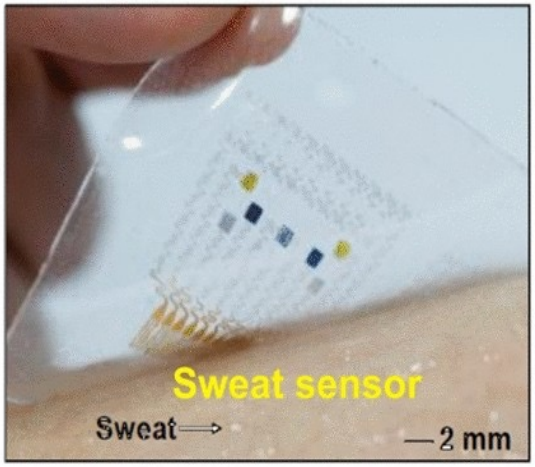






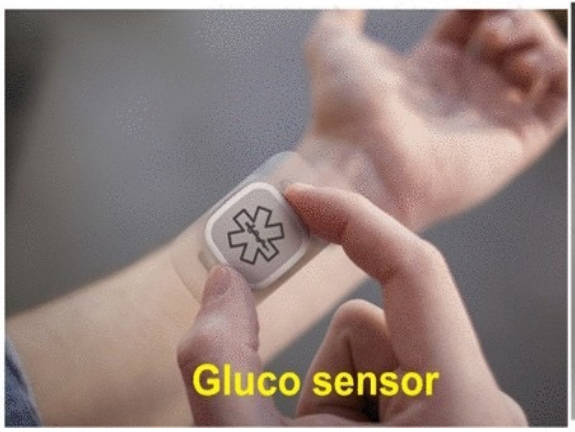
The Microstrip (patch) Antenna

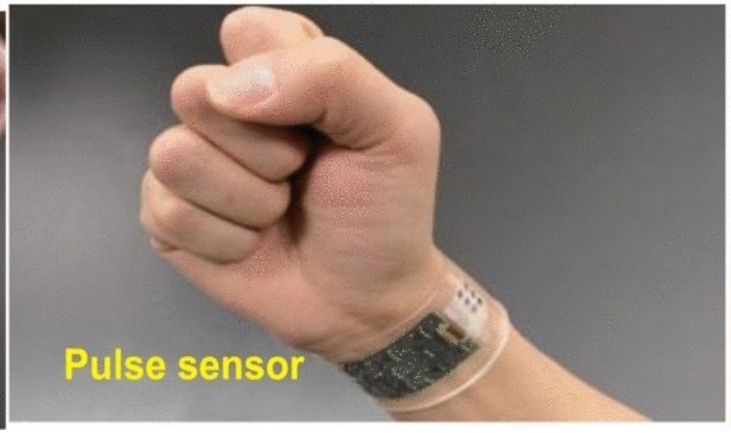






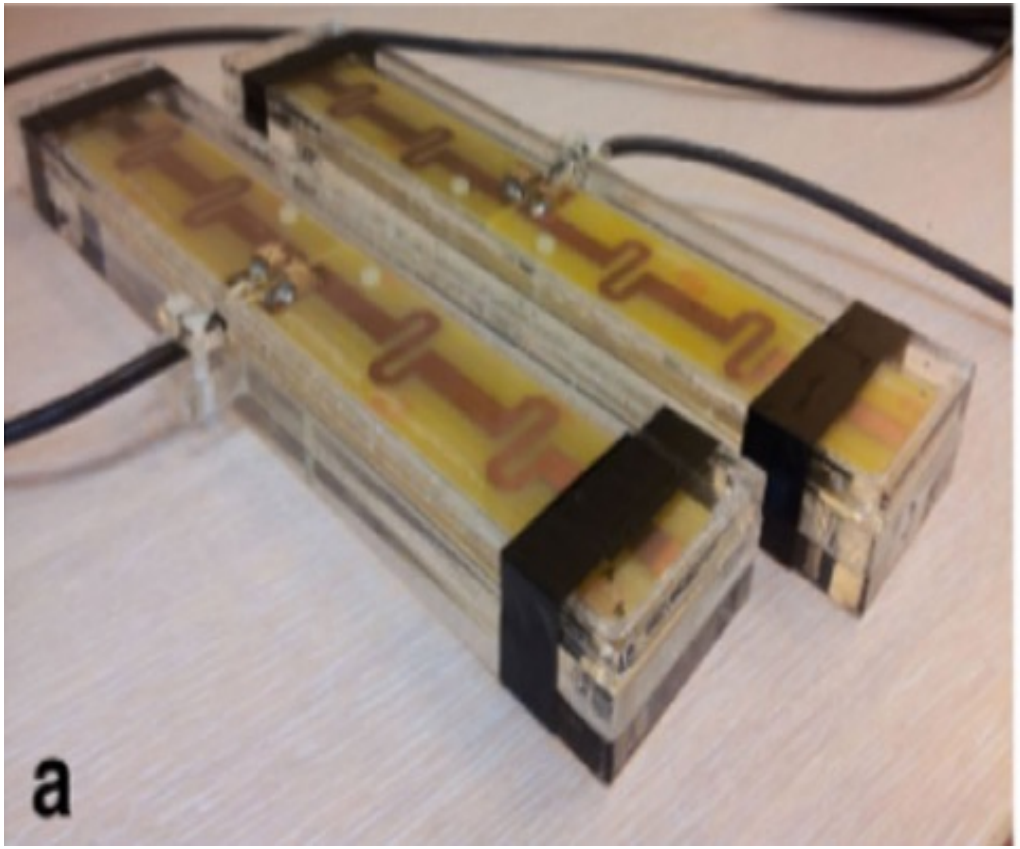


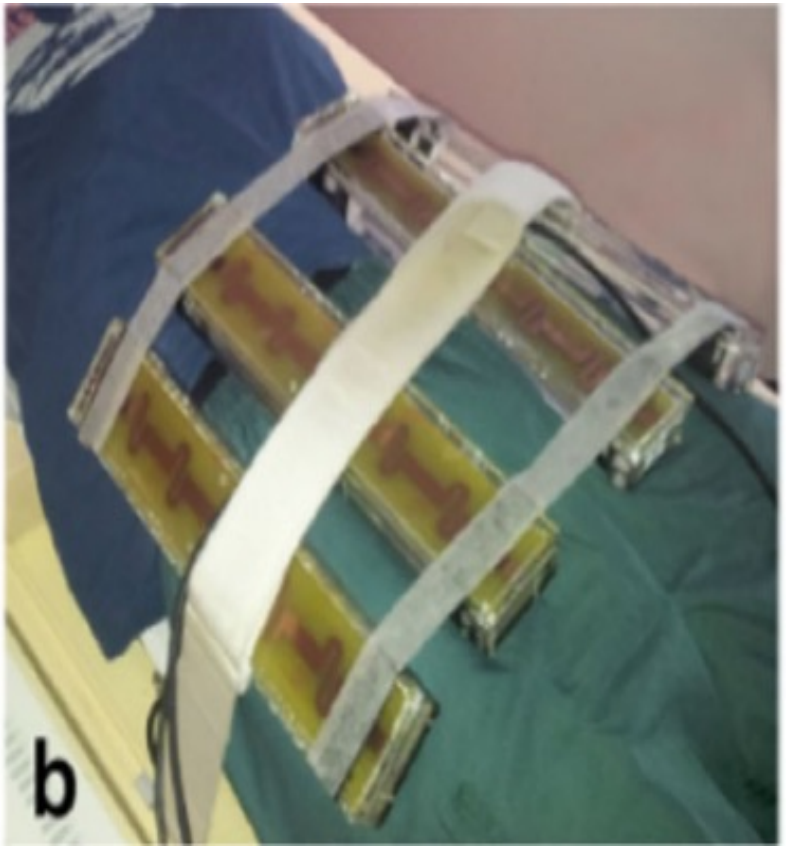




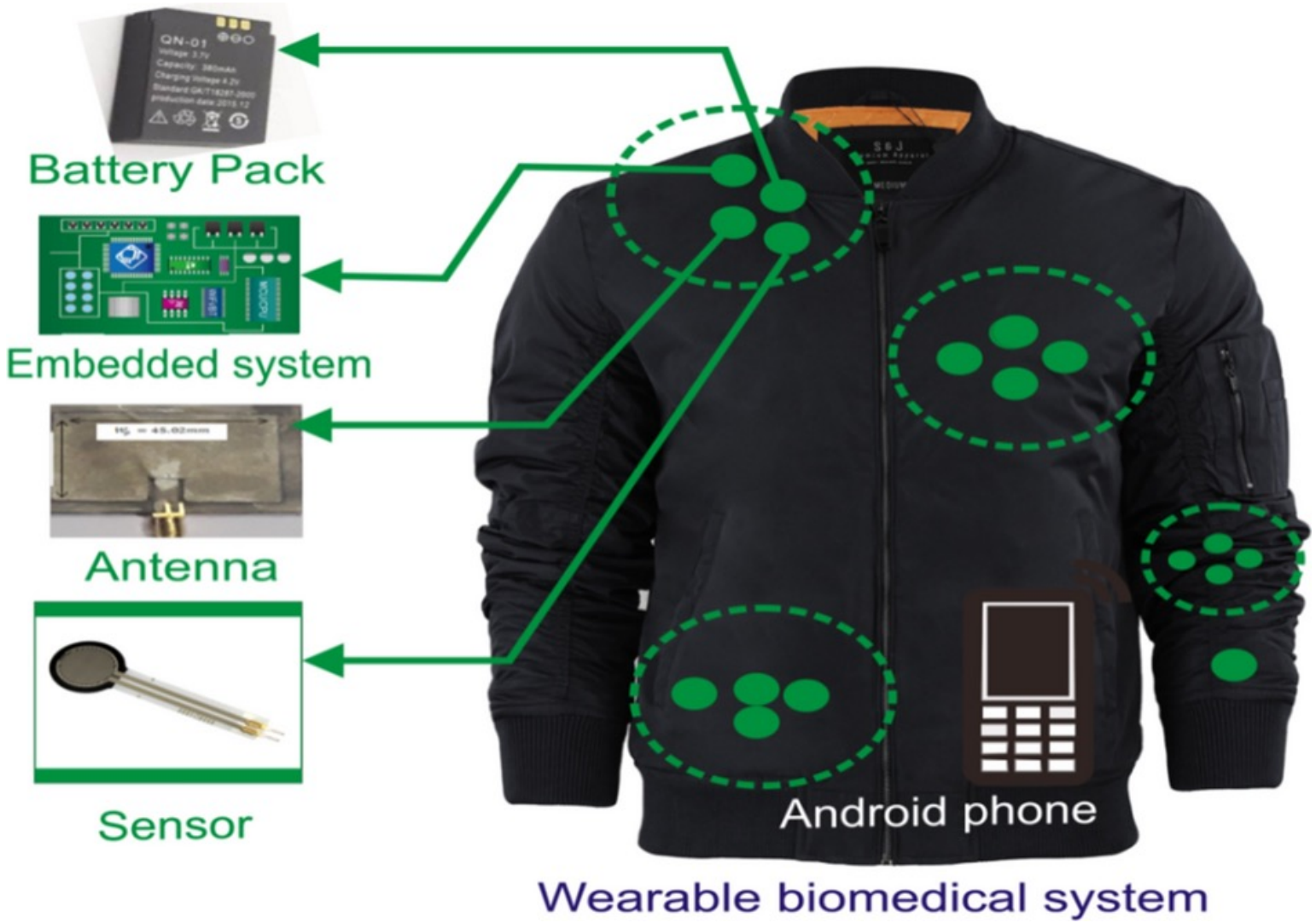


The Microstrip (patch) Antenna

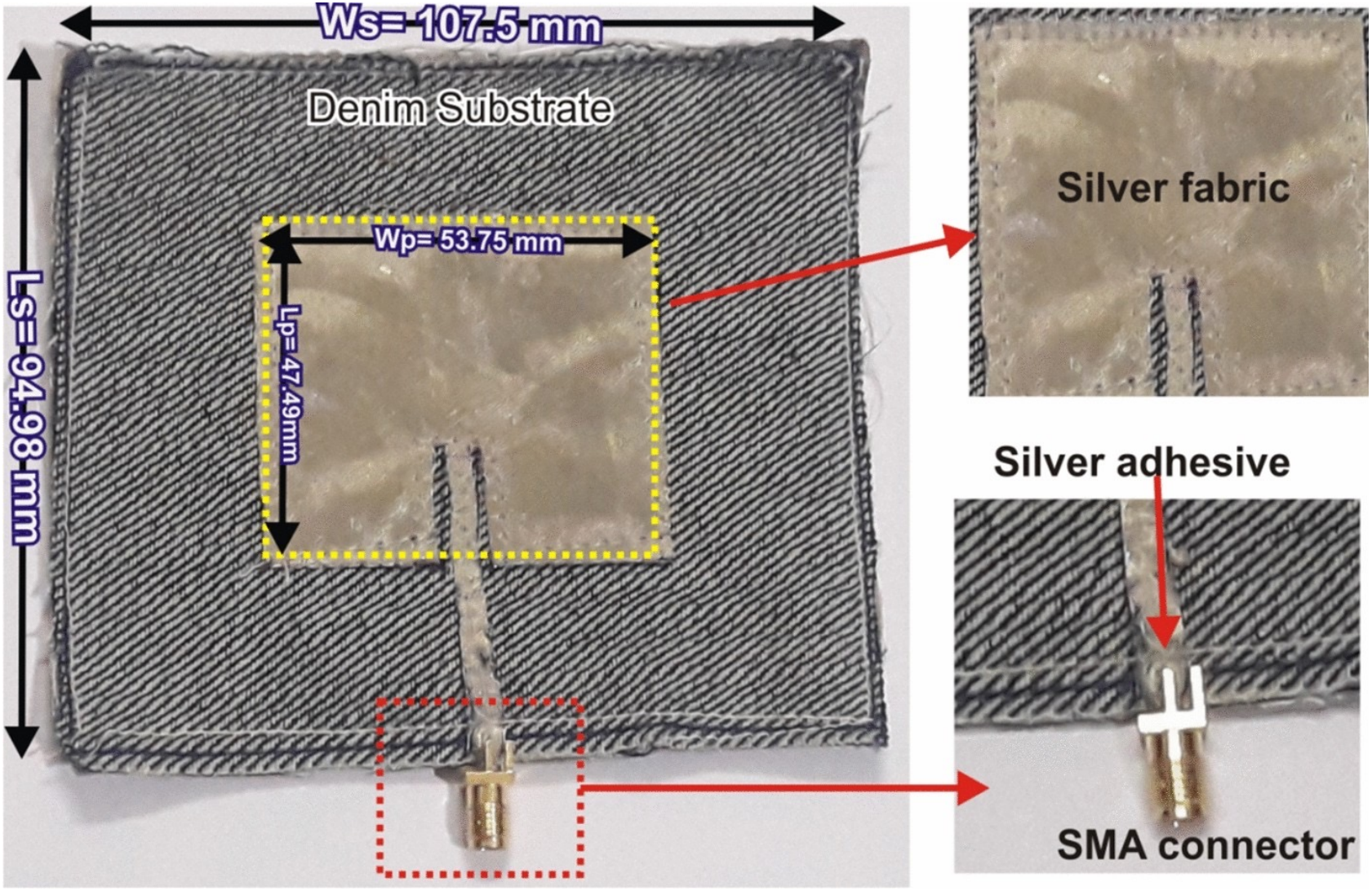




The Microstrip (patch) Antenna

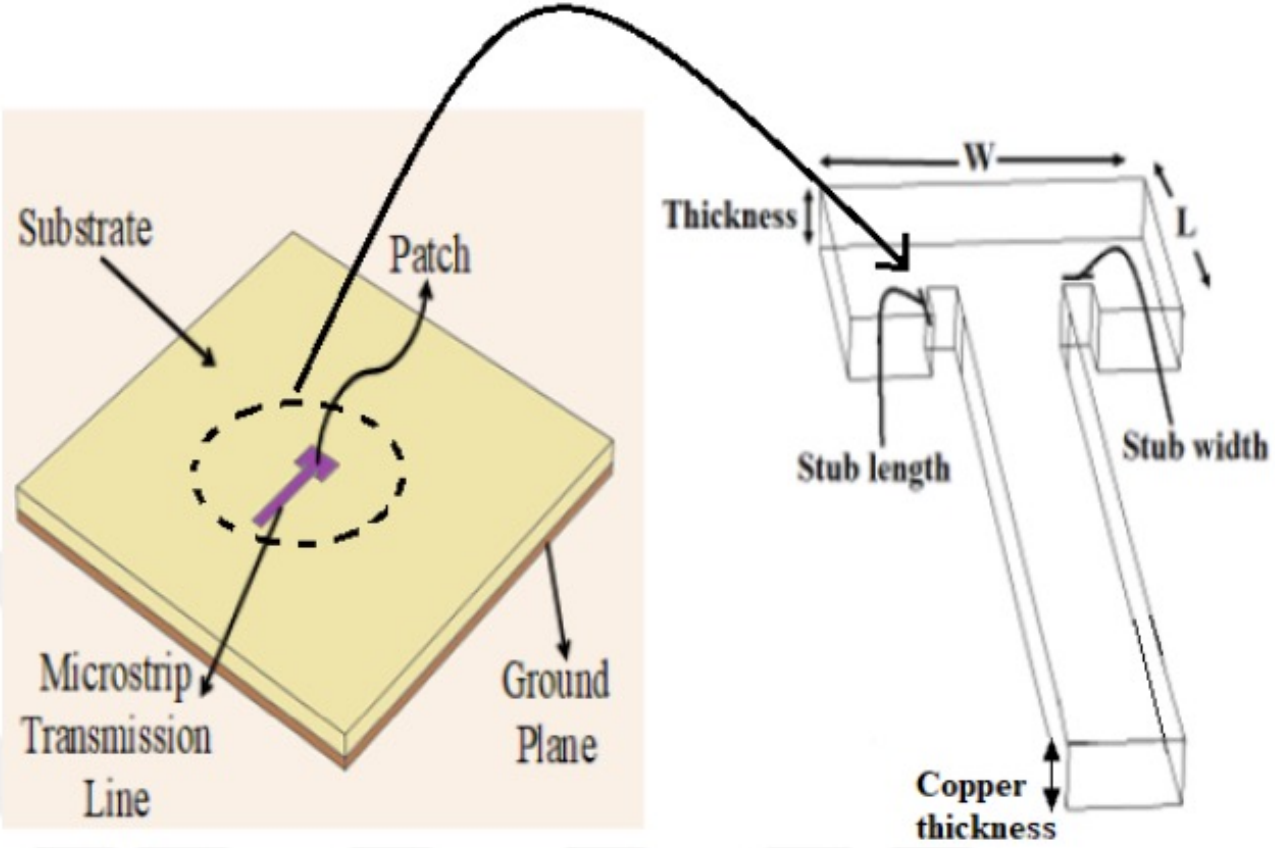


The Microstrip (patch) Antenna



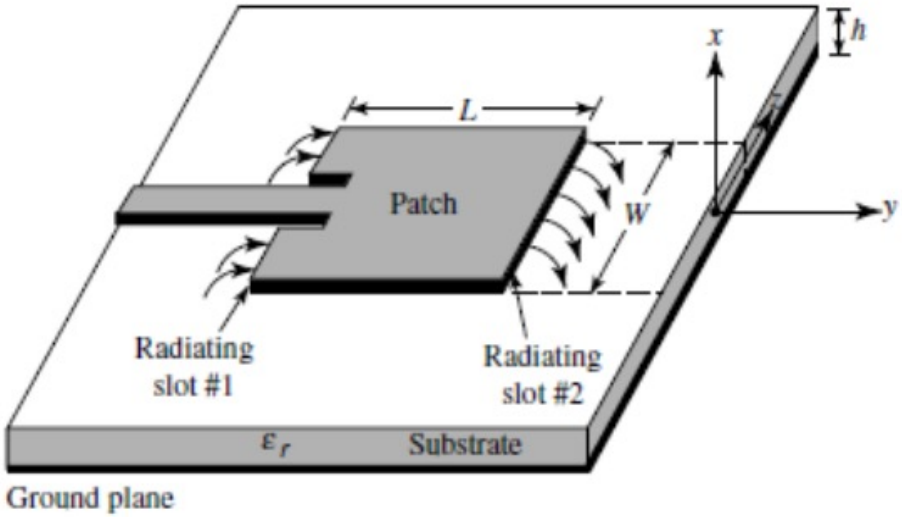
The Microstrip (patch) Antenna

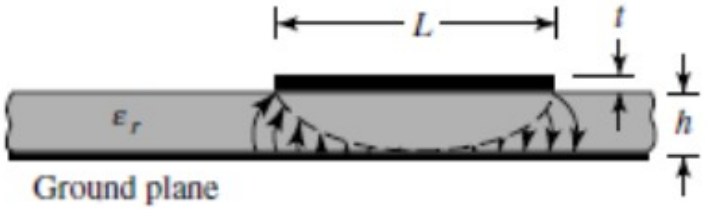
The most important parameters for resonance are the length, substrate thickness and substrate material. Other parameters have minor effects.

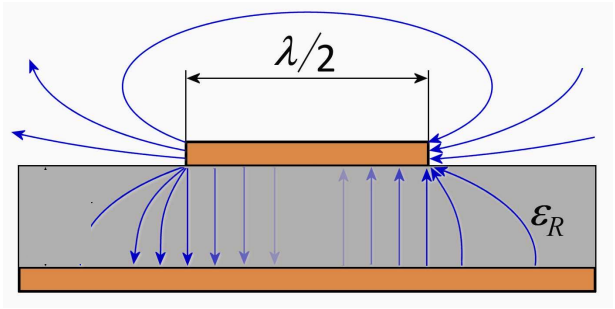


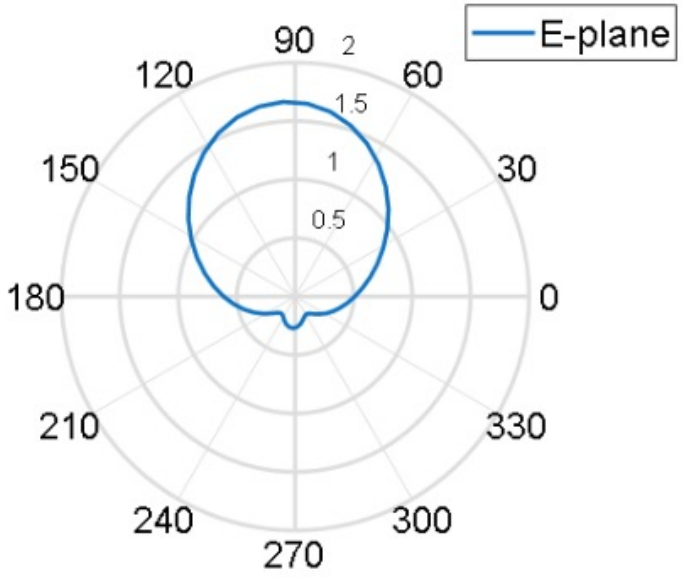
The Microstrip (patch) Antenna

The principle of operation is the radiation of "fringing fields" within the antenna "cavity". These field form a radiation pattern that is maximum above the antenna and minimum below the antenna.



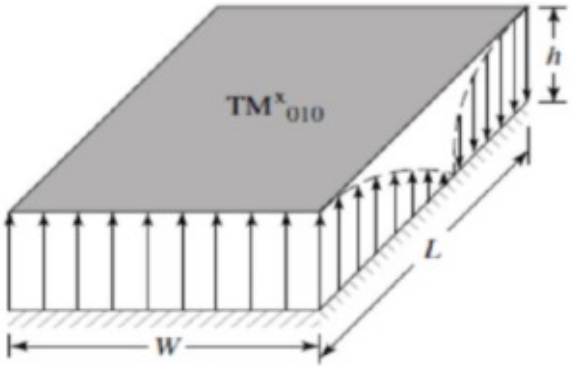




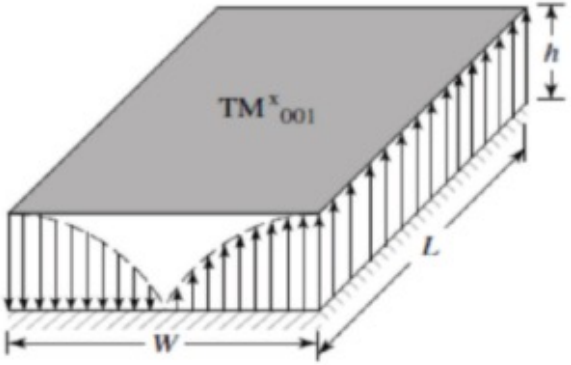


The Microstrip (patch) Antenna

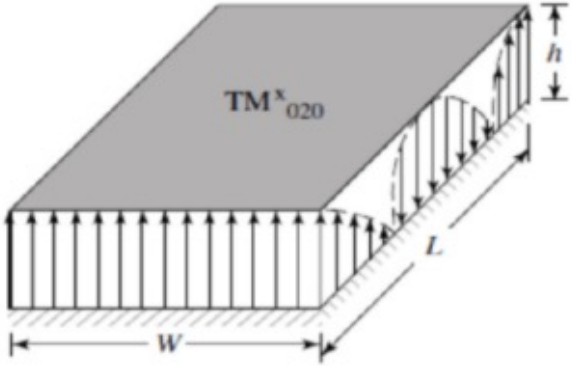
Like all other resonating structures, it is possible to excite different modes of the antenna cavity. This possibility is used to produce "circular" polarization.



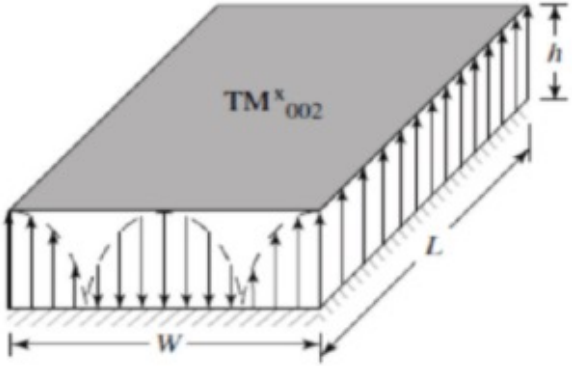
(a) TM^x_{010}



(b) TM^x_{001}



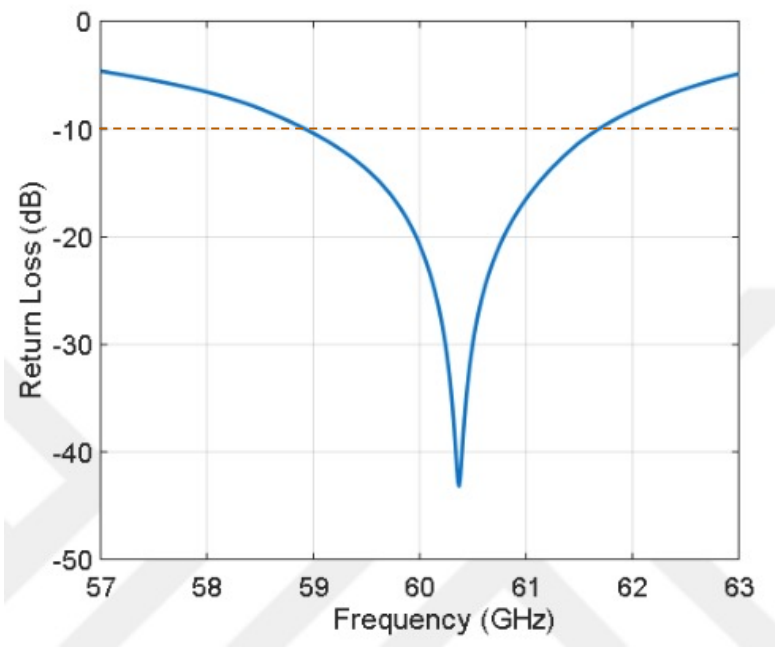
(c) TM^x_{020}

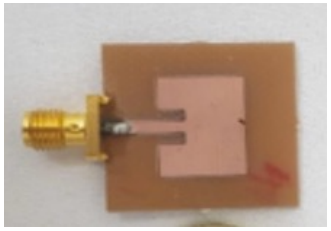


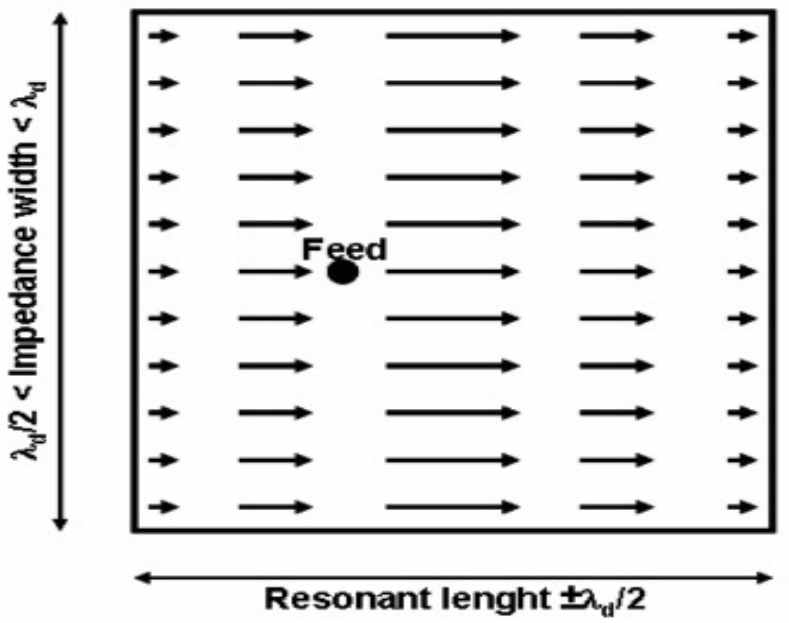
(d) TM^x_{002}

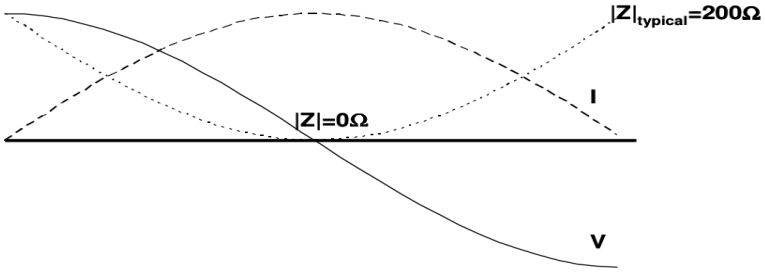
The Microstrip (patch) Antenna

The patch antenna is characterized by many parameters (like all antennas): resonance frequency, bandwidth (1-3%), radiation pattern (not omnidirectional), gain (6-9 dBi), directivity(5-7 dB), polarization (linear- circular), etc.



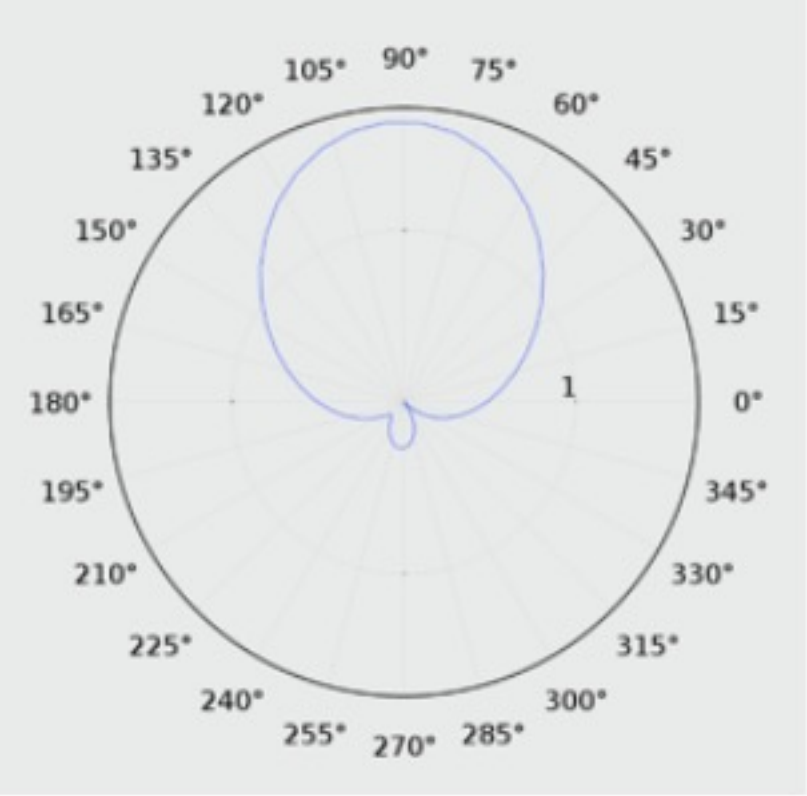


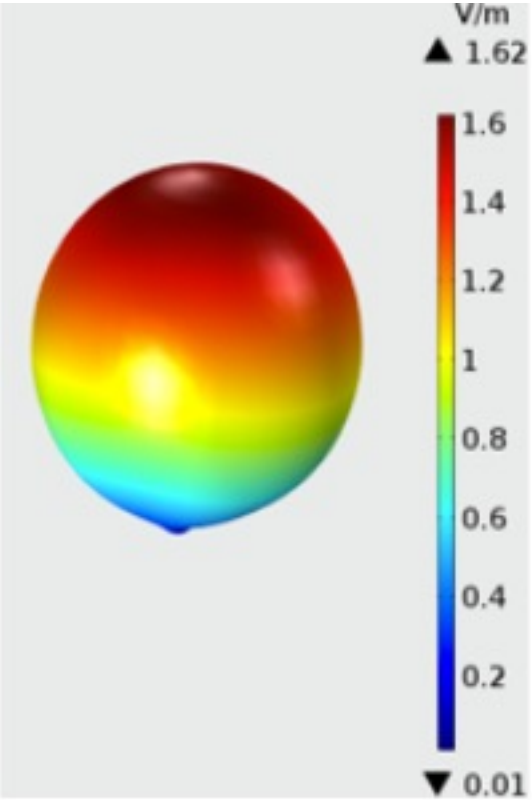




The Microstrip (patch) Antenna

The patch antenna is characterized by many parameters (like all antennas): resonance frequency, bandwidth (1-3%), radiation pattern (not omnidirectional), gain (6-9 dBi), directivity (5-7 dB), polarization (linear- circular), etc.





Q & A

Send me an email if you have any questions.